

Nithish Reddy Mendu

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EDUCATION

Mahatma Gandhi Institute of Technology (MGIT) — Hyderabad, India
Bachelor of Technology, Computer Science and Engineering **CGPA: 9.3/10** 2024 – 2028

TECHNICAL SKILLS

Languages	C++, Java, Python, JavaScript, SQL
Frontend	React.js, HTML5, CSS3, Bootstrap, Zustand, React DnD
Backend	Node.js, Express.js, REST APIs, Socket.IO, WebSockets
Desktop Development	Electron.js
Databases	MongoDB, SQLite, SQLite FTS5, MySQL, Redis
AI & Search	Tesseract OCR, Transformer Embeddings, Cosine Similarity Search
Tools	Git, GitHub, Postman, VS Code, Cloudinary, Stripe, BullMQ, Vercel, Render
Core CS	Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

PROJECTS

ElectricSearch | Electron.js, React.js, Node.js, SQLite (FTS5), Tesseract OCR, Transformers, Worker Threads [GitHub](#) | [Live Demo](#)

- Developed an offline-first desktop search engine using Electron.js that extracts text from screenshots using native Tesseract OCR and indexes **1,000+ images** in SQLite FTS5, enabling **8–12 ms** full-text search.
- Engineered a **multi-threaded OCR pipeline** using Node.js Worker Threads, automatic folder monitoring, and background indexing, improving OCR throughput by **56%** while maintaining a responsive UI.
- Implemented offline semantic search using **Xenova/bge-small-en-v1.5**, generating **384-dimensional vector embeddings** and retrieving contextually relevant results in **40–60 ms** using cosine similarity without relying on cloud APIs.
- Designed a secure Electron architecture using **25+ IPC APIs**, preload scripts, repository pattern, and context isolation to safely coordinate the React UI, filesystem, OCR engine, AI pipeline, and SQLite database.

TeamFlow | React.js, Zustand, React DnD, Node.js, Express.js, MongoDB, Socket.IO, JWT, Cloudinary, Redis, BullMQ [GitHub](#) | [Live Demo](#)

- Developed a Trello-inspired real-time collaboration platform exposing **40+ REST APIs** supporting authentication, boards, lists, tasks, invitations, media uploads, notifications, and team collaboration.
- Designed a transaction-safe backend using **MongoDB multi-document transactions**, role-based access control, and cascade deletion across multiple collections to maintain data consistency.
- Built real-time collaboration using **Socket.IO**, implementing board synchronization, direct messaging, typing indicators, read receipts, notifications, and **10+ synchronized socket events**.
- Developed a modular React frontend using **custom hooks**, **Zustand** for state management, and **drag-and-drop** interactions; implemented async email delivery via **BullMQ** and **Redis** background workers.

LocalLynk | React.js, Node.js, Express.js, MongoDB, JWT, Stripe API [GitHub](#) | [Live Demo](#)

- Developed a full-stack hyperlocal marketplace enabling users to buy and sell products within a configurable radius using **MongoDB 2dsphere indexes** for efficient location-based discovery.
- Built secure authentication using **JWT**, Google OAuth, protected routes, user profile management, and authorization workflows.
- Implemented **15+ REST APIs** supporting products, orders, users, payments, and buyer–seller communication while integrating **Stripe** for secure online transactions.
- Designed responsive React interfaces featuring advanced search, category filtering, distance-based sorting, inventory management, and complete order lifecycle management.

ACHIEVEMENTS & CERTIFICATIONS

- Solved **440+ Data Structures & Algorithms** problems across **LeetCode (250+)** and **GeeksforGeeks (190+)**.
- Runner-up**, 24-Hour Hackathon, KL University among **120+ participating teams**.
- Certifications: **Postman API Fundamentals**; Data Science Master Program (NumPy & Pandas) – CampusX.